

Younghun Lee

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Profile

Computer Science Ph.D. candidate interested in **natural language processing** and its real world applications, specifically in subjectivity analysis, representation learning, explainable AI, and social network analysis.

Education

Aug 2019 to Present	Purdue University , West Lafayette, Indiana Doctor of Philosophy in Computer Science • Study Emphasis: Natural Language Processing and Computational Social Science • Research: Subjective ground of individuals, advised by Dr. Dan Goldwasser • Award: Graduate Teaching Award in 2023.
Sep 2015 to Apr 2017	University of Michigan , Ann Arbor, Michigan Master of Science in Information • Study Emphases: Information Retrieval and Human-computer Interaction • Research: Interaction retrieval and analysis, advised by Dr. Mark W. Newman • Award: Outstanding Graduate Student Instructor of the Year in 2016-2017.
Mar 2008 to Feb 2015	Seoul National University , Seoul, South Korea Bachelor of Science in Electrical and Computer Engineering • Study Emphases: Applied Electrical Engineering and Information Visualization • Research: Data mining and visualization, advised by Dr. Joonhwan Lee • Award: Dean of Faculty of Liberal Education Prize (1st Prize) in Fall 2012.

Selected Publications

Under review	Younghun Lee and Dan Goldwasser, <i><u>Towards Characterizing Subjectivity of Individuals via Counterfactual Retrieval-Augmented Generation</u></i>
Jul 2026	Younghun Lee , Amir Bralin, Nobel Sanjay Rebello and Dan Goldwasser, <i><u>Toward Just-in-Time Adaptive Feedback: Enhancing Student Learning via Knowledge-Grounded LLM</u></i> , ACL 2026 Workshop (BEA 2026), USA
Nov 2025	Younghun Lee and Dan Goldwasser, <i><u>SOLAR: Towards Characterizing Subjectivity of Individuals through Modeling Value Conflicts and Trade-offs</u></i> , EMNLP 2025, China
July 2024	Younghun Lee , Sungchul Kim, Ryan Rossi, Tong Yu, Xiang Chen, <i><u>Learning to Reduce: Towards Improving Performance of Large Language Models on Structured Data</u></i> , ICML 2024 Workshop on Long-Context Foundation Models
Mar 2024	Younghun Lee , Dan Goldwasser, Laura Schwab-Reese, <i><u>Towards Understanding Counseling Conversations: Domain Knowledge and Large Language Models</u></i> , Findings of EACL 2024, Malta
Jul 2023	Tianqi Wang, Lei Chen, Xiaodan Zhu, Younghun Lee , Jing Gao, <i><u>Weighted Contrastive Learning With False Negative Control to Help Long-tailed Product Classification</u></i> , ACL 2023 Industry Track, Canada
Dec 2022	Younghun Lee and Dan Goldwasser, <i><u>Towards Explaining Subjective Ground of Individuals on Social Media</u></i> , Findings of EMNLP 2022, UAE
Jul 2020	Hou Wei Chou, Younghun Lee , Lei Chen, Yandi Xia, Wei-Te Chen, <i><u>CamemBERT and BiT Feature Extraction for Multimodal Product Classification and Retrieval</u></i> , SIGIR eCom, China
Oct 2018	Younghun Lee , Seunghyun Yoon and Kyomin Jung, <i><u>Comparative Studies of Detecting Abusive Language on Twitter</u></i> , EMNLP 2018 Workshop on Abusive Language Online, Belgium

Professional Experience

Jun 2024 to Aug 2024	Research Intern at LG AI Research , Seoul, Korea Conducted research on AI safety, specifically on crawling and annotating real-world data that guides expert LLMs generate safe and faithful answers given a query
Jun 2023 to Aug 2023	Research Scientist/Engineer Intern at Adobe , San Jose, California - Conducted research on utilizing large language models to better perform on structured data QA tasks, mainly on table QA datasets - Training a policy language model to reduce the input table context helps LLMs infer more accurately
Jun 2020 to Aug 2020	Research Science Intern at Rakuten Institute of Technology , Boston, Massachusetts - Conducted research on long-tailed item categorization in natural language processing, improving the state-of-the-art models using unlabelled data augmentation - Participated in SIGIR eCom'20 Data Challenge for multi-modal and cross-modal product classification and achieved top 5 performance
Apr 2018 to Dec 2018	Research Intern at Seoul National University, Machine Intelligence Lab Conducted independent research on computational social science and natural language processing, focusing on abusive language detection on Twitter
Jun 2016 to Aug 2016	UX Developer Intern at Gale , Farmington, Michigan Developed a web application for internal salespeople to help them visualize Gale's online learning resources and demonstrate the eLearning framework

Selected Projects

Feb 2017 to Apr 2017	Revealing the Network and Diffusion Patterns of False Information - Customize <i>Twitter API for Python</i> that retrieves millions of messages containing specific URL information - Model friends and follower network graphs, analyze with centrality measures, and identify characteristics of false information
Nov 2016 to Dec 2016	Investigating the Effect of Presidential Election Results on Hate Crimes - Develop a custom Twitter message retrieval program on hate crimes - Investigate the statistical correlation between election results and hate crime related sentiment on Twitter
Nov 2016 to Apr 2017	User Interaction Pattern Retrieval and Analysis - Implement an Android web emulator from a Java application that captures user interaction patterns in mobile applications and stores log data in Firebase database - Recruit 100 users from Amazon Mechanical Turk and analyze log files
May 2014 to Jun 2014	Sawtooth Bubble Chart—Visualizing Human Emotion (Summary Paper) - Retrieve and categorize Twitter data by time, age, and geolocation - Implement a model that visualizes arousal and valence level of human emotion of each tweet using Affective Norms for English Words dictionary
Dec 2013 to Jan 2014	Visualizing Articles with Different Viewpoints Develop a web application that compares viewpoints of two Korean news articles describing the same event, and visualize their similarities and differences with word clouds

Reference

Dan Goldwasser	Associate Professor - Purdue University, Department of Computer Science - Email: dgoldwas@purdue.edu
Moontae Lee	Assistant Professor - University of Illinois, at Chicago - Email: moontae@uic.edu